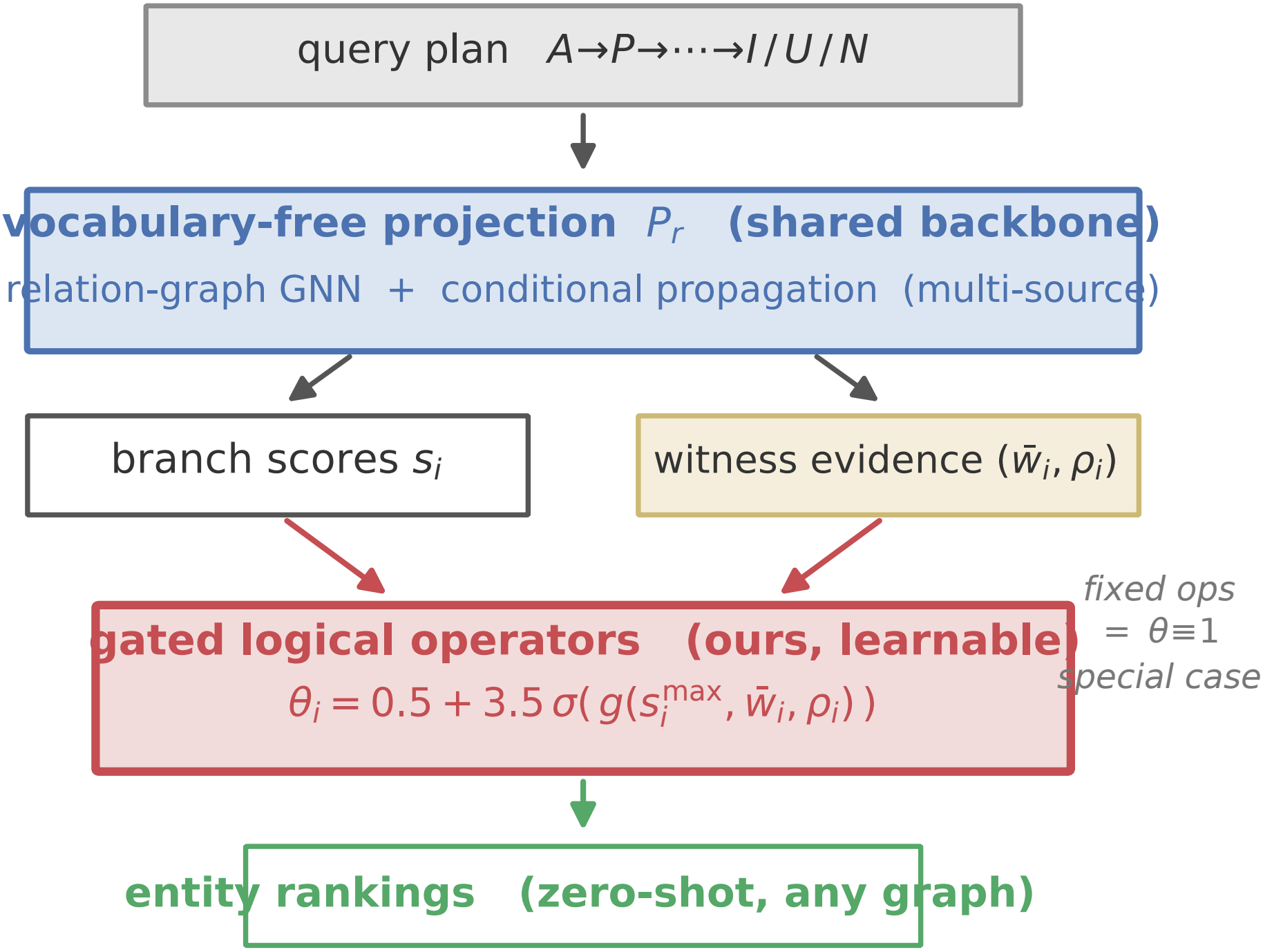
query plan $A \to P \to \cdots \to I \,/\, U \,/\, N$
vocabulary-free projection P_r (shared backbone)
relation-graph GNN + conditional propagation (multi-source)
branch scores s_i
witness evidence $(\bar{w}_i, \rho_i)$
gated logical operators (ours, learnable)
$\theta_i = 0.5 + 3.5\,\sigma(\, g(s_i^{\max}, \bar{w}_i, \rho_i)\,)$
fixed ops
$= \theta \equiv 1$
special case
entity rankings (zero-shot, any graph)